Product Requirements Document

Students' Assignment and Lecture Note Portal System | Focused MVP and selective reuse of existing ICP

Ibadan City Polytechnic, Ibadan | Planning baseline: 24 October 2026

1. Product summary

Project title: Development of a Students' Assignment and Lecture Note Portal System for Ibadan City Polytechnic, Ibadan.

Product: A standalone, browser-based portal through which approved lecturers publish course lecture notes and assignments; eligible students access course content, submit coursework, and view published assignment feedback; administrators manage access and course allocations.

Scope guard: This is not a course registration, semester examination result, CGPA, payment, attendance, or full learning management system. Reuse only the proven ICP foundation needed for this project.

2. Confirmed existing ICP baseline and reuse policy

Verified from connected services: Cloudflare Pages project icp-course-portal with production URL https://icp-course-portal.pages.dev/ and successful deployment reported on 7 October 2026. Supabase project ICP Course Registration and Result Processing System, reference qazayzyrwuwlendhfald, is active. Verified database structures include profiles, user_roles (student/lecturer/admin), students, lecturers, departments, programmes, academic_sessions, semesters, courses, programme_courses, course_allocations and course_registrations. Existing project has an active create-admin function and profile-pictures storage bucket.

Evidence limit: Full deployed frontend source, CSS tokens, typography and exact hex color values were not retrievable in this inspection. This document deliberately specifies an exact source-derived visual copy, not unverified colors or a proposed new theme. Database existence does not prove every UI journey works.

Element	Reuse decision	Condition
Branding, colors, spacing, fonts, logo, icons	COPY EXACTLY	Extract actual assets and style definitions from existing ICP source, never infer
Login, recovery, role-based route patterns	REUSE/ADAPT	Check behavior and security before copying
Header, sidebar, cards, dashboard layouts, responsive styles	REUSE	Remove irrelevant pages and controls
Profile, department, course, lecturer assignment data model	ADAPT	Recreate minimal subset in separate database
Course registration, result batches, GPA/CGPA, grades ledger	EXCLUDE	Do not port tables, RPCs, navigation or workflows
Assignment upload, note download, submissions, grading	BUILD NEW	Only core features of the project

3. Users and permissions

Role	Permissions
Admin	Approve/manage user access; create/manage departments, course records, student-course membership and lecturer-course allocation; oversee counts and content metadata; suspend accounts. No implicit access to private submission files beyond explicit administrative oversight policy.
Lecturer	See allocated courses; upload/replace/archive own lecture notes; create/edit/publish course assignments; set close time and marks; review relevant student submissions; record and publish scores and written feedback.
Student	See assigned courses; view/download visible notes and assignments; upload/replace submission before deadline as configured; see own submission state, published marks and feedback.

4. Functional requirements

FR-01 Account access

• Secure email/password authentication; password reset; session expiry and role-based landing screen.

• Admins approve or provision lecturers and students; users cannot self-assign admin or lecturer permissions.

• Profile shows name, role, department/programme where relevant; only admins manage lecturer-course assignments.

FR-02 Course visibility

• Course catalogue and department/programme mappings maintained by admin.

• Lecturer sees only allocated courses; students see only courses to which admin has assigned them.

• Course pages organize materials and assignments by course; optionally tag academic session/semester as metadata, with no registration workflow.

FR-03 Lecture notes

• Lecturer uploads PDF, DOCX or PPTX (configurable supported types), file title, short description and course.

• Students can list, preview PDFs when browser supports it, and download published notes; uploads are protected.

• Lecturer can replace, unpublish or delete own notes; admin can manage inappropriate content.

FR-04 Assignments

• Lecturer creates title, instructions, due date/time, maximum mark, course and optional attachment; draft/publish/close states.

• Students see instructions, attachments, due dates, status and submission history for their own work.

• After due time, system marks work late/closed according to a clearly configured policy; default: submission closes at deadline.

FR-05 Student submissions

• Student uploads PDF, DOCX or allowed file with optional comment; validates type/size and ownership.

• Show submission timestamp and successful confirmation; default one active submission per student per assignment, replaceable until due time.

• Lecturer sees course roster, submitted/not submitted and downloadable submissions; private submissions are not publicly shareable.

FR-06 Assignment grading and feedback

• Assigned lecturer enters numeric score between zero and maximum mark, optional written feedback, and draft/published state.

• Students only see their own published scores and feedback.

• Assignment marks are standalone coursework feedback; no CGPA, final result publication or transcript integration.

FR-07 Basic notifications and reports

• In-app notices for new published notes, assignments, nearing deadlines and published feedback; email/push are explicitly out of MVP.

• Admin sees simple counts; lecturer sees submitted/pending totals and downloadable/exportable CSV summary if straightforward.

• Avoid workflow engines, SMS, WhatsApp bots and background scheduling unless separately approved.

5. Main screens

Admin	Lecturer	Student
Dashboard / overview	Dashboard / allocated courses	Dashboard / enrolled courses
Users and approval	Course notes list/upload	Lecture notes list/download
Departments, courses, allocations	Assignments create/edit	Assignments list/details
Basic reports	Submissions and grading	My submissions and feedback

6. Suggested minimal data model (fresh Supabase project)

Recreate only necessary tables, adapting proven ICP conventions where appropriate. Proposed tables: profiles, user_roles, departments, courses, course_memberships (student-course), lecturer_allocations, lecture_notes, assignments, submissions, assignment_feedback, notifications. Use Supabase Auth for identities, private storage buckets for course material and student submissions, and a publicly readable logo only when intended.

Use immutable UUID identifiers, foreign keys, created_at/updated_at timestamps, explicit ownership, status checks, server-side deadline checks, unique (assignment_id,student_id) active-submission rule, and sensible cascade/restrict behavior. Avoid copying existing users or production records into a demonstration system; create distinct test accounts and synthetic sample content.

7. Non-functional requirements and security

• Visual fidelity: existing ICP colors, logos, typography, spacing, component styles, breakpoints, interaction conventions and dashboard shells reproduced exactly from actual source assets. No invented gradients, altered palette or new design language.

• Responsive: desktop, tablet and small Android screen; accessible focus/keyboard states; sensible empty/loading/error states.

• Security: RLS on every exposed table; storage access rules enforce course membership and lecturer allocation; privileged keys never shipped to browser; safe file types and size limits; no public

student-submission URLs.

• Reliability: clear upload validation; duplicate-submit protection; deadline enforcement using database/server time; recoverable errors.

• Isolation: entirely separate Git repository (or branch extracted into new repository), Supabase project, Cloudflare Pages project and credentials; do not modify ICP production.

• Maintainability: concise migrations, small readable modules, clear README, environment template and setup notes.

8. Acceptance criteria / demo script

• Admin logs in, creates course or uses seeded course, assigns one lecturer and two students.

• Lecturer logs in, publishes one lecture note; assigned student sees and downloads it, unassigned user is denied.

• Lecturer publishes assignment with deadline and maximum score; student submits permitted file and sees time and status.

• Lecturer opens submission, records score/feedback and publishes; only that student sees their own result.

• Second student submission/grade remains private; nonallocated lecturer cannot access records.

• Mobile UI matches original ICP design tokens and works without horizontal overflow.

• Screens for course registration, CGPA and semester final results are absent.

• Existing ICP website and database remain unchanged after development and deployment.

9. Out of scope and change control

Explicitly exclude admission processing, course registration, examination results, GPA/CGPA, timetables, attendance, fee payment, bulk institutional imports, parent portals, plagiarism detection, live classrooms, video calls, quizzes, messaging/chat, AI grading, WhatsApp/SMS integration, and complex notification queues. Any addition requires separate written approval.

10. Dependencies, unknowns and delivery

• Get read-only access to existing ICP frontend repository or source archive to identify the real CSS, fonts, assets, layout and component behavior; preserve exact design.

• Confirm limits for uploads (proposed defaults: 15 MB notes; 10 MB submissions) and institutional assignment policy during implementation; these are defaults, not verified ICP rules.

• Deliver: frontend, database SQL migrations, RLS/storage policies, seed/demo accounts instructions, tests, README, Cloudflare setup and screenshots.

• Do not claim pixel-perfect completion until visual comparison against existing ICP screenshots at matching desktop/mobile viewports.